

Project Overview and Navigation Guide

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This document provides a high-level architectural map of the project, detailing our current position, the utility of recent deliverables, and the required next steps.

1. The Big Picture: Where Are We?

The core objective of this project is to **utilize an LLM to read 28 scientific papers and automatically extract CMOCs (Context - Mechanism - Outcome Configurations) matching the human-derived results of the Richmond (2020) review.**

To achieve this, the project is structured into three distinct phases:

- **Phase 1: Zero-shot Extraction:** Feeding 28 papers to the AI for initial, unguided data extraction. *(Status: Complete. Output stored in `evidence_table.json`).*
- **Phase 2: Evaluation Framework & HITL (Human-in-the-loop):** Building a measurement system to benchmark the AI against the human standard (Richmond), and establishing a mechanism for human oversight. *(Status: Complete. This encompasses our most recent work).*
- **Phase 3: Refinement:** Utilizing the established metrics to iteratively restructure prompts and architecture, forcing the AI to achieve higher accuracy. *(Status: Pending).*

Current Position: We have successfully completed **Phase 2**.

2. Purpose of Recent Deliverables

The scripts and output files generated recently serve three critical functions for your PhD dissertation:

1. The Evaluator (`Phase1_Relationship_Evaluation.md`)

- *Purpose:* To replace vague claims of "good AI performance" with rigorous scientific metrics. This script evaluates the AI against the exact Richmond R01-R40 golden standard.
- *Result:* The current baseline F1 score is **8%**. This low initial score is expected for a zero-shot run, and it demonstrates to your academic committee that we have a highly transparent and strict measurement tool in place.

2. The Adjudication Template (`per_paper_adjudication_template.csv`)

- *Purpose:* The 8% score is a global metric. The professor requires granular knowledge of exactly *where* the AI failed in each paper. This Excel-compatible file aggregates all 57 CMOC extractions.
- *Utility:* It pre-fills the AI's responses and leaves blank columns for **you (the**

researcher) to manually grade and correct the outputs.

3. The Human-in-the-Loop Audit Logs (HITL_Audit_Log.md)

- *Purpose:* To address the professor's concern about "blind AI execution." This log proves that the system pauses at critical milestones, awaiting human review and approval before proceeding.
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3. How to Verify the Work

To verify the integrity and optimization of the current system, you do not need to review the underlying code. You act as the **Project Validator** by reviewing the following outputs:

1. **Review outputs/Phase1_Relationship_Evaluation.md:** This file provides a clear report on F1 Scores, Precision, Recall, and a detailed error analysis showing exactly which relationships the AI missed or falsely predicted compared to Richmond.
2. **Review outputs/per_paper_adjudication_template.csv:** Open this in Excel. You will see 57 rows (representing 57 CMOCs). Spot-check a few rows to assess if the AI's extracted Context, Intervention, and Outcome are logically sound, and note the blank human_corrected columns waiting for your input.
3. **Review outputs/Professor_Handoff_Summary.md:** Read this academic summary designed specifically for Professor Zheng. It directly and accurately addresses the concerns raised during your last meeting.

If these files display correct scientific logic and transparent grading mechanisms, it confirms the underlying codebase is fundamentally sound.

4. Next Action Steps

The workflow now temporarily shifts from automated AI execution to manual researcher review.

Your Immediate Tasks:

1. Open outputs/per_paper_adjudication_template.csv using Excel.
2. Utilize the golden standard data (or your domain expertise) to evaluate the AI's extractions row by row.
3. Wherever the AI has made an error, input the correct answer into the corresponding human_corrected_... columns (e.g., human_corrected_context, human_corrected_triples).
4. Save the completed Excel file.

Subsequent Phase: Once you complete this manual grading, we will possess a perfect "Per-Paper Golden Standard." We will then re-engage the AI to begin **Phase 3 (Refinement)**: restructuring prompts and system architecture to drive the F1 score from **8%** to a publishable threshold (e.g., 70-80%).

